in the MLP activation space and specifies a linear combination of neurons. Since the rows of A are neuron activations, each MLP feature $\mathbf{z}_i$ captures a co-activation pattern across neurons, so multiplying by W_V maps it to the residual stream defining a *residual stream feature* $\mathbf{f}_i \in \mathbb{R}^d$:

$$\mathbf{f}_i = W_V \mathbf{z}_i = \sum_{j=1}^{d_a} z_{i,j} \mathbf{v}_j \tag{4}$$

We encourage these linear combinations to be *sparse*, as typically observed in MLP layers in LLMs (Geva et al., 2021) (see details below). The *coefficient matrix* Y is restricted to nonnegative values, which specify how each input representation $\mathbf{h}_i$ is reconstructed using the MLP features. Each entry $Y_{i,j}$ indicates how strongly MLP feature i contributes to sample j. Together, Y and Z provide a parts-based decomposition, where the original activations are reconstructed as additive combinations of the MLP features.

Optimization To learn the decomposition, we initialize the matrices Z and Y such that the entries of Z are drawn from a uniform distribution $\mathcal{U}(0,1)$, and the entries of Y are drawn from a standard normal distribution $\mathcal{N}(0,1)$. Then we use the Multiplicative Updates scheme of Ding et al. (2010), which alternates between a closed-form update of the matrix Z and a multiplicative update of the nonnegative matrix Y in order to minimize the Frobenius norm between the matrix A and its reconstruction ZY. Lastly, we impose sparsity by applying a hard winner-take-all (WTA) operator to every column of Z, keeping only the largest $p\%$ of entries (by absolute value) and zeroing the remainder as suggested in Peharz and Pernkopf (2012). For full details, see §A.

4 Concept Detection Evaluations

We start by evaluating if the features learned by SNMF capture generalizable input-dependent patterns. We show that the MLP features consistently activate in response to concept-related inputs and remain inactive for neutral inputs, suggesting that a stable group of neurons underlies each concept. These features perform comparably to those identified by SAEs trained on the MLP output and better than equally sized SAEs trained on the same data, supporting their relevance to the model's input.

4.1 Experiment

We follow previous work (Paulo et al., 2024; Gur-Arieh et al., 2025; Wu et al., 2025) and measure the ability of MLP features to discern between contexts that exemplify the associated concept and those that do not. Concretely, we first describe the concept captured by an MLP feature based on inputs that activate it, and then evaluate if the feature activates on inputs that were generated based on the description.

To describe the concept captured by MLP features, we leverage GPT-4o-mini (OpenAI et al., 2024) and prompt it to describe prominent patterns in the inputs that most strongly activate it (Bills et al., 2023; Bricken et al., 2023; Choi et al., 2024; Paulo et al., 2024) (see §G.2 for the prompt). Then, for every description we generate five activating sentences that express the concept and five neutral sentences that do not. Next, we run each sentence through the model and record the MLP activations for every token during the forward pass.

To evaluate how well an MLP feature aligns with its description, we compute the cosine similarity between the feature and each token activation in a sentence, and take the maximum similarity per sentence.[3] We then average these maximum scores across the activating sentences to obtain $\bar{a}_{\text{activating}}$, and across the neutral sentences to obtain $\bar{a}_{\text{neutral}}$. Lastly, we measure the log-ratio between the mean scores for the activating and neutral sentences to obtain the Concept Detection score (S_{CD}):

$$S_{CD} := \log \frac{\bar{a}_{\text{activating}}}{\bar{a}_{\text{neutral}}} \tag{5}$$

We use logarithm to normalize the scale for comparability across methods. A positive log-ratio indicates greater similarity with activating sentences, where larger values indicate better separability.

We apply this evaluation to SNMF-learned MLP features with $k = 100, 200, 300, 400$ on multiple layers of Gemma-2-2B (Team et al., 2024), Llama-3.1-8B (Grattafiori et al., 2024), and GPT-2 Small (Radford et al., 2019). Additional details on the training of SNMF are in §D. As a baseline, we use publicly available SAEs (He et al., 2024; Lieberum et al., 2024; Gao et al., 2025) trained on MLP outputs (SAE out), i.e. $W_V\mathbf{a}$ (Equation 3), and SAEs trained on MLP activations (SAE act) with the same dataset and number of features as SNMF

[3] We use cosine similarity as opposed to projection to make the scales of different methods comparable (see details in §B).

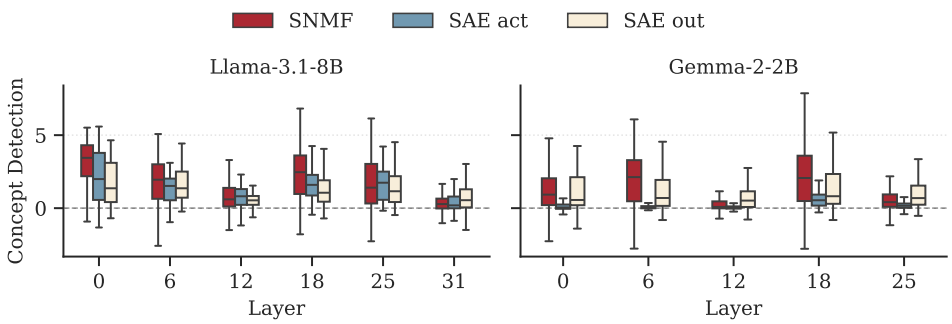

Figure 2: The distribution of concept detection scores across selected layers in Llama-3.1-8B (left) and Gemma-2-2B (right) for SNMF (k=100), SAE trained on MLP activations with the same dataset as SNMF (SAE act) and publicly-available SAEs trained on MLP outputs (SAE out).

(see details in §C). For SNMF, we generate concept labels by analyzing the input samples that most strongly activate it, using the coefficient matrix Y. For SAE act, we take the feature descriptions from Neuronpedia, which are derived using autointerp (Paulo et al., 2024) and similarly rely on the highest-activating examples. For SAEs trained on the SNMF dataset, we generate descriptions based on their top-activating contexts using the same prompts used for generating descriptions for SNMF.

4.2 Results

Figure 2 shows the Concept Detection score for SNMF MLP features, SAE act MLP features and SAE out features in Llama-3.1-8B and Gemma-2-2B. For SNMF we present the results for $k = 100$, which shows similar trends to other values of k. Additional results showing similar trends for GPT-2 Small and comparisons across different k values are provided in §D. Across different layers and models, most of the SNMF MLP features (>75%) obtain a positive concept detection score, indicating that the majority of SNMF-discovered features activate more for concept-related inputs than for neutral ones. This suggests that the majority of MLP features are meaningfully tied to the appearance of specific concepts in the input. Interestingly, we observe comparatively high scores in the first layer of both Llama-3.1-8B and GPT-2. We hypothesize that this is because the activation representations have passed through fewer attention layers so they encode fewer entangled concepts, making them easier to decompose.

When comparing methods, SNMF generally achieves concept detection scores that are comparable to or exceed those of existing SAEs trained on MLP outputs and a better performance compared to SAEs trained on the SNMF dataset. A minority of MLP features receive negative scores, indicating they respond more strongly to neutral contexts, possibly reflecting sensitivity to noise or layer-specific biases. The lower scores of SAE act may partly reflect challenges in achieving sparsity with less features and smaller datasets. For Gemma in particular, training was less stable as we observed that increased sparsity often led to a large number of dead features. We believe that with additional tuning and broader parameter searches, smaller SAEs have the potential to perform more competitively than the versions evaluated here.

These results show that SNMF reliably identifies sets of neurons that correspond to human-interpretable concepts. We note that SNMF's higher concept detection scores may stem from the improved interpretability of SNMF provided by the coefficient matrix Y, which gives a straightforward association between features and their most activating contexts. In Table 1, we present examples of identified concepts along with the tokens most strongly associated with them according to the coefficient matrix Y.

5 Concept Steering Evaluations

In this section, we turn to measure the steering effectiveness of MLP features while ensuring that model fluency is preserved. We show that MLP features match and often exceed the steering performance of strong unsupervised and supervised baselines trained on MLP outputs. Together with the interpretability results, these findings show the success of our approach in isolating meaningful neuron combinations used by the model.

Layer	Concept	Top Activating Contexts
0	The word "resonate" and variations	captivating blend of nostalgia and innovation that resonates Pont Mirabeau captivates the eye, resonating with the a new advertising campaign that would resonate more strongly
12	Actions related to implementing or establishing	the police enacted curfews and employed sound cannons In JavaScript, establishing a coherent code layout schools must implement a comprehensive framework that supports the evaluation
23	Historical documentation	stone carvings, striving to reveal the profound stories etched by generations the rich historical narratives preserved by the ancients forgotten texts found in the dusty archives

Table 1: Example concepts discovered by SNMF across different layers of Llama-3.1-8B, along with their top activating contexts extracted from the coefficient matrix Y.

5.1 Experiment

We evaluate whether MLP features discovered by SNMF causally influence the model's output in a concept-consistent manner (Paulo et al., 2024; Gur-Arieh et al., 2025; Wu et al., 2025). We benchmark SNMF against SAE out, SAE act, and a strong supervised baseline, Difference-in-Means (DiffMeans) (Wu et al., 2025; Rimsky et al., 2024; Marks and Tegmark, 2024; Turner et al., 2024).

We feed the prompt "<BOS> I think that" to the model while steering with a feature during inference. To control the steering's strength, we perform a grid search over seven target values of KL divergence between the output logits from a standard forward pass and from an intervened pass where we amplify the feature (Paulo et al., 2024). As some features may suppress a concept rather than promote it, we consider both the feature $\mathbf{f}$ and its negation $-\mathbf{f}$. For each KL divergence value and sign, we sample 8 generations, to obtain an overall of 112 completions.

Next, we prompt GPT-4o-mini (OpenAI et al., 2024) to score each generated continuation along two axes: *concept expression* (concept score) and *fluency* (fluency score).[4] We use the evaluation prompts by Wu et al. (2025), where each axis is scored from 0-2. For each feature, we take the best result over KL/sign combinations. We report two scores: (a) the concept score, which evaluates how well the generated completions align with the target concept, and (b) the harmonic mean of the concept and fluency scores, which quantifies how effectively a feature can steer the model's generation toward the target concept while preserving output coherence. We exclude fluency scores from the main figure as they are less informative in isolation. Fluency scores are provided in §E.

Concept scores are computed with respect to a reference feature description. Gur-Arieh et al. (2025) has shown that input-centric descriptions are better for describing features in early layers, while output-centric descriptions are better for describing features in upper layers. Therefore, we generate descriptions for early layer features using activating inputs (as explained in §4). For later layers, across all methods, we use an output-centric description. For an MLP feature (SNMF and SAE act), $\mathbf{z}_i$, we calculate its projection onto the residual stream via the MLP out weights,

$$\mathbf{f}_i := W_V \mathbf{z}_i \tag{6}$$

For features of SAE out and DiffMeans we directly use $\mathbf{f}_i$ as they were trained on MLP outputs. Next, we project the feature to the model's vocabulary space using the unembedding matrix, and generate the description based on the set of tokens corresponding to the top and bottom logits in the projection (Gur-Arieh et al., 2025).

We apply this evaluation to SNMF features with $k = 100$ across multiple layers in LLaMA-3.1–8B and Gemma-2–2B. For SAE act we use all 100 features. For SAE out, we randomly sample 100 features from Gemma Scope and LLaMA Scope and use their corresponding decoder rows. Since DiffMeans is supervised, we apply it to learn features for the concepts identified by SNMF and SAEs. For each concept, we use its description to generate 72 activating and 72 neutral sentences (as done in §4) and define the feature as the difference between the average token representations for each set (Wu et al., 2025).

5.2 Results

Figure 3 presents the results, showing varying performance across the layers of both models. Early

[4] We omit the instruct score used in Wu et al. (2025) as we evaluate base models rather than instruction tuned.